Module 2 Assignment

Student Record Manager

Java Program

```
import java.io.*;
import java.util.*;
import java.util.regex.*;

class Student {
    int id;
    String name;
    String email;
    String course;

    Student(int id, String name, String email, String course) {
        this.id = id;
        this.name = name;
        this.email = email;
        this.course = course;
    }

    @Override
    public String toString() {
        return id + "," + name + "," + email + "," + course;
    }
}

public class StudentRecordManager {

    static final String FILE_NAME = "students.txt";
    static Scanner sc = new Scanner(System.in);

    // Email validation using Regex
    public static boolean isValidEmail(String email) {
        String emailRegex = "^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";
        return Pattern.matches(emailRegex, email);
    }

    // Add student
    public static void addStudent() {
        try {
            System.out.print("Enter Student ID: ");
            int id = Integer.parseInt(sc.nextLine());

            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Email: ");
            String email = sc.nextLine();

            if (!isValidEmail(email)) {
                throw new IllegalArgumentException("Invalid email format!");
            }

            System.out.print("Enter Course/Branch: ");
            String course = sc.nextLine();

            Student student = new Student(id, name, email, course);

            // Save student to file
            FileWriter fw = new FileWriter(FILE_NAME, true);
            BufferedWriter bw = new BufferedWriter(fw);

            bw.write(student.toString());
            bw.newLine();

            bw.close();

            System.out.println("Student added successfully!");

        } catch (NumberFormatException e) {
            System.out.println("Invalid Student ID! Please enter a number.");
        } catch (IllegalArgumentException e) {
            System.out.println(e.getMessage());
        } catch (IOException e) {
            System.out.println("Error while saving student data.");
```

```
        }
    }

    // Read student data
    public static void readStudents() {
        try {
            File file = new File(FILE_NAME);

            if (!file.exists()) {
                System.out.println("No student records found.");
                return;
            }

            BufferedReader br = new BufferedReader(new FileReader(file));

            String line;

            System.out.println("\n----- Student Records -----");

            while ((line = br.readLine()) != null) {
                String[] data = line.split(",");

                System.out.println("Student ID : " + data[0]);
                System.out.println("Name       : " + data[1]);
                System.out.println("Email      : " + data[2]);
                System.out.println("Course     : " + data[3]);
                System.out.println("---------------------------");
            }

            br.close();

        } catch (IOException e) {
            System.out.println("Error while reading student data.");
        }
    }

    // Main method
    public static void main(String[] args) {

        while (true) {

            System.out.println("\n===== Student Record Manager =====");
            System.out.println("1. Add Student");
            System.out.println("2. Read Student Data");
            System.out.println("3. Exit");
            System.out.print("Enter your choice: ");

            try {
                int choice = Integer.parseInt(sc.nextLine());

                switch (choice) {
                    case 1:
                        addStudent();
                        break;

                    case 2:
                        readStudents();
                        break;

                    case 3:
                        System.out.println("Exiting program...");
                        sc.close();
                        return;

                    default:
                        throw new IllegalArgumentException(
                                "Invalid choice! Please select 1, 2 or 3.");
                }

            } catch (NumberFormatException e) {
                System.out.println("Invalid input! Please enter a number.");
            } catch (IllegalArgumentException e) {
                System.out.println(e.getMessage());
            }
        }
    }
}
```